```
import pandas as pd                     # For data handling
import seaborn as sns                  # For built-in dataset + visualization
import matplotlib.pyplot as plt        # For plotting graphs
```

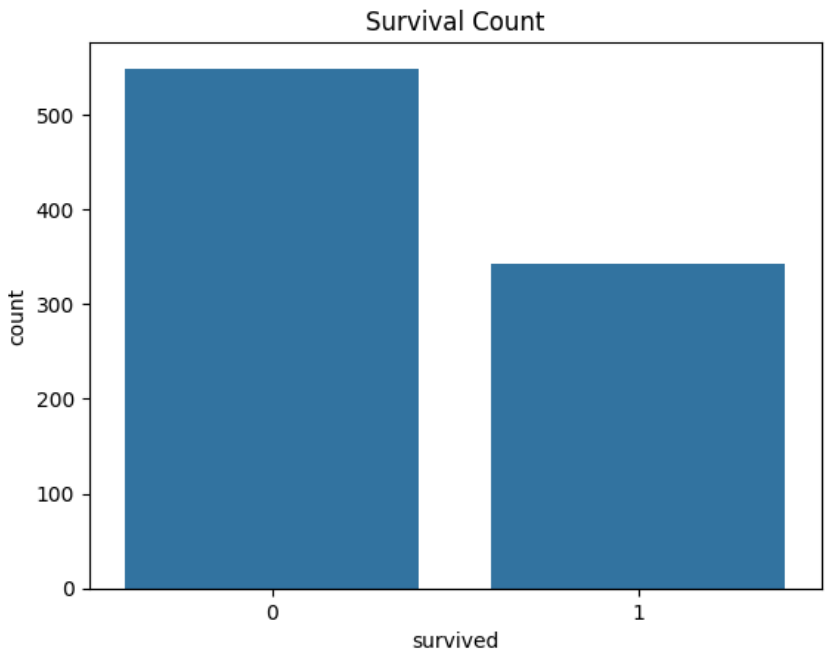
```
df = sns.load_dataset('titanic')   # Load dataset directly from seaborn
```

```
print(df.head())     # Show first 5 rows
print(df.info())     # Show structure and data types
print(df.describe()) # Statistical summary
```

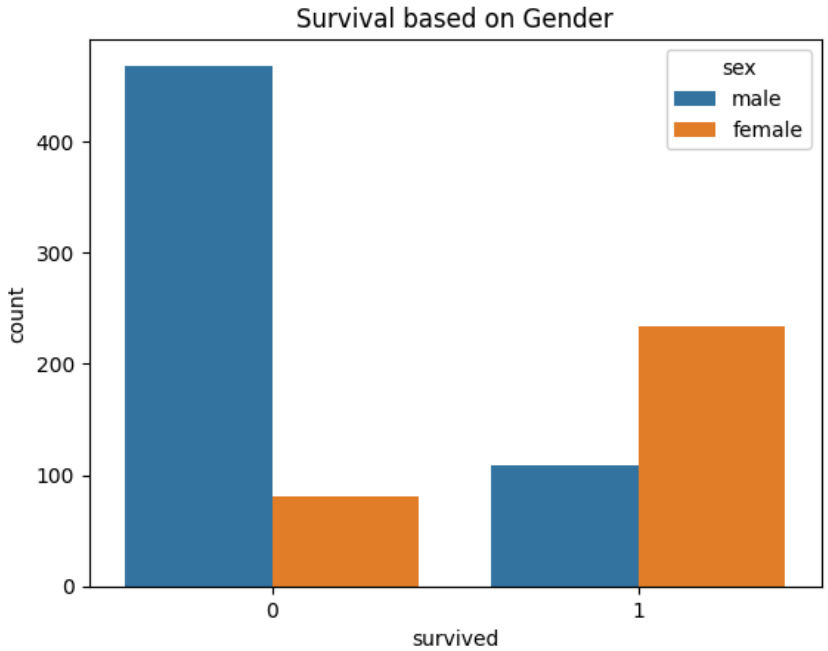
```
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None
         survived      pclass         age       sibsp       parch        fare
count  891.000000  891.000000  714.000000  891.000000  891.000000  891.000000
mean     0.383838    2.308642   29.699118    0.523008    0.381594   32.204208
std      0.486592    0.836071   14.526497    1.102743    0.806057   49.693429
min      0.000000    1.000000    0.420000    0.000000    0.000000    0.000000
```

```
25%      0.000000    2.000000   20.125000    0.000000    0.000000    7.910400
50%      0.000000    3.000000   28.000000    0.000000    0.000000   14.454200
75%      1.000000    3.000000   38.000000    1.000000    0.000000   31.000000
max      1.000000    3.000000   80.000000    8.000000    6.000000  512.329200
```

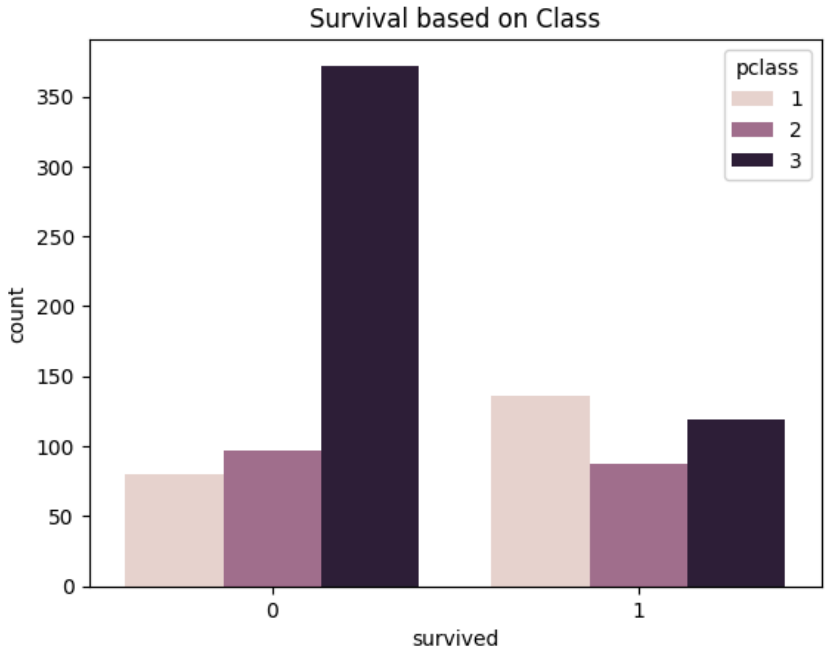
```
sns.countplot(x='survived', data=df) # Helps identify imbalance
plt.title("Survival Count")
plt.show()
```



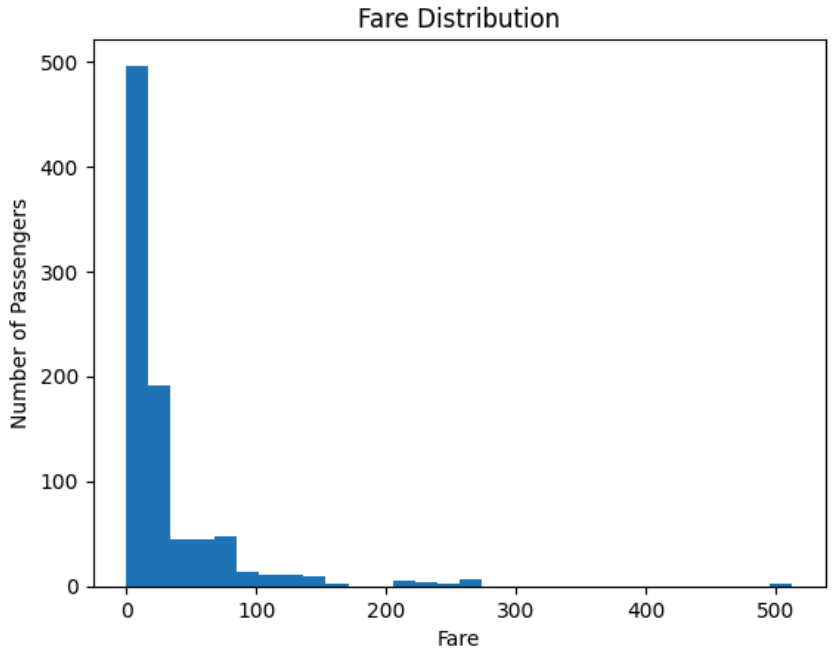

```
sns.countplot(x='survived', hue='sex', data=df)
plt.title("Survival based on Gender")
plt.show()
```



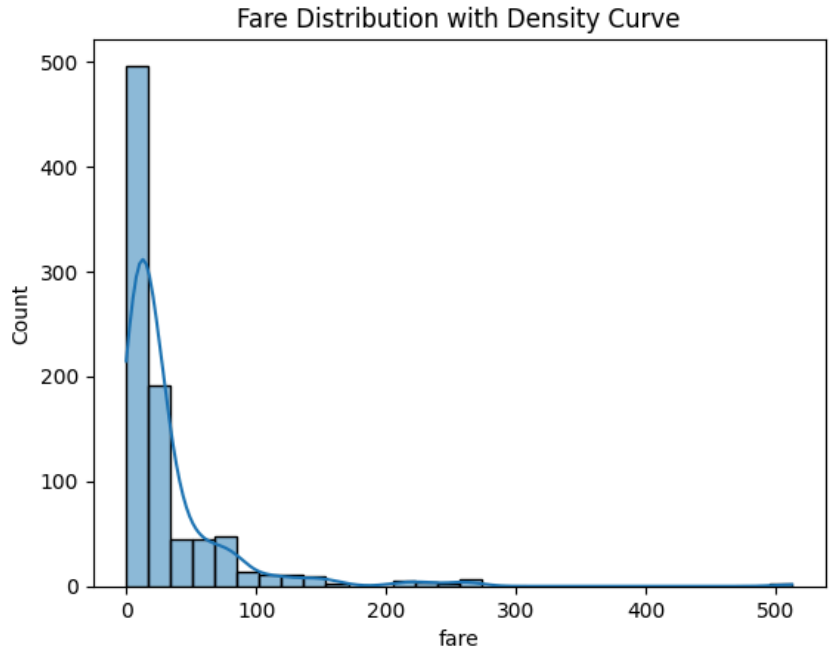

```
sns.countplot(x='survived', hue='pclass', data=df)
plt.title("Survival based on Class")
plt.show()
```




```
plt.hist(df['fare'], bins=30)   # Plot histogram
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Number of Passengers")
plt.show()
```




```
sns.histplot(df['fare'], bins=30, kde=True)
plt.title("Fare Distribution with Density Curve")
plt.show()
```



2. How did you load the dataset?

Using sns.load_dataset('titanic')

3. Why use Seaborn instead of pandas?

Seaborn provides built-in datasets and better visualization tools.

4. What is a countplot?

A plot showing frequency/count of categorical data.

5. What is a histogram?

Graph showing distribution of numerical data.

6. What does bins mean?

Number of intervals used to divide data.

7. What is KDE in histogram?

Kernel Density Estimation → smooth curve representing distribution.

8. What pattern did you observe in survival?

Females had higher survival rate than males.

9. How does passenger class affect survival?

Higher class passengers had better survival chances.

10. What type of distribution is fare?

Right-skewed distribution.

11. Why is fare skewed?

Few passengers paid very high fares, majority paid low.

12. Difference between plt.hist() and sns.histplot()?

plt.hist() sns.histplot()

Basic plotting Advanced visualization

No KDE Supports KDE

13. Why use visualization?

To identify patterns, trends, and relationships in data.

14. What is EDA (Exploratory Data Analysis)?

Process of analyzing data using statistics and visualization.

15. Can histogram be used for categorical data?

No, only for numerical data.

16. What happens if bins are too large or too small?

Too large → loss of detail

Too small → noisy graph

17. Why use hue in seaborn?

To compare categories within the same plot.

"Using seaborn, patterns in Titanic dataset were analyzed through countplots and fare distribution was visualized using histogram, showing a right-skewed distribution."

1. Histogram

Used for:

Distribution of numerical data

Example:

Titanic fare

Why:

Shows how values are spread

Helps detect skewness

2. Count Plot / Bar Plot

Used for:

Categorical data

Example:

Survival count

Gender distribution

Why:

Shows frequency/count of categories

3. Box Plot

Used for:

Detect outliers

Compare distributions

Example:

Fare vs Class

Why:

Shows median, quartiles, extreme values

4. Scatter Plot

Used for:

Relationship between two numerical variables

Example:

Age vs Fare

Why:

Shows correlation (positive/negative)

5. Line Plot

Used for:

Time-based data

Example:

Sales over time

Why:

Shows trends clearly

6. Heatmap

Used for:

Showing correlation between variables

Example:

sns.heatmap(df.corr(), annot=True)

Why:

Quickly shows relationships between features

Data Type Plot

Numerical (1 variable) Histogram

Categorical Countplot

Numerical vs Numerical Scatter

Numerical vs Category Boxplot

Time-based Line plot

Multiple variables Heatmap

1. Why use histogram instead of bar chart?

Histogram is for continuous data, bar chart is for categorical

2. When would you use a scatter plot?

When analyzing relationship between two numerical variables

3. What does a box plot show?

Median, quartiles, and outliers

4. What is skewness?

Measure of asymmetry in distribution

5. Why use heatmap?

To visualize correlation between variables

6. What happens if wrong plot is used?

Misleading interpretation of data

7. Why use hue in seaborn plots?

To compare categories within same graph

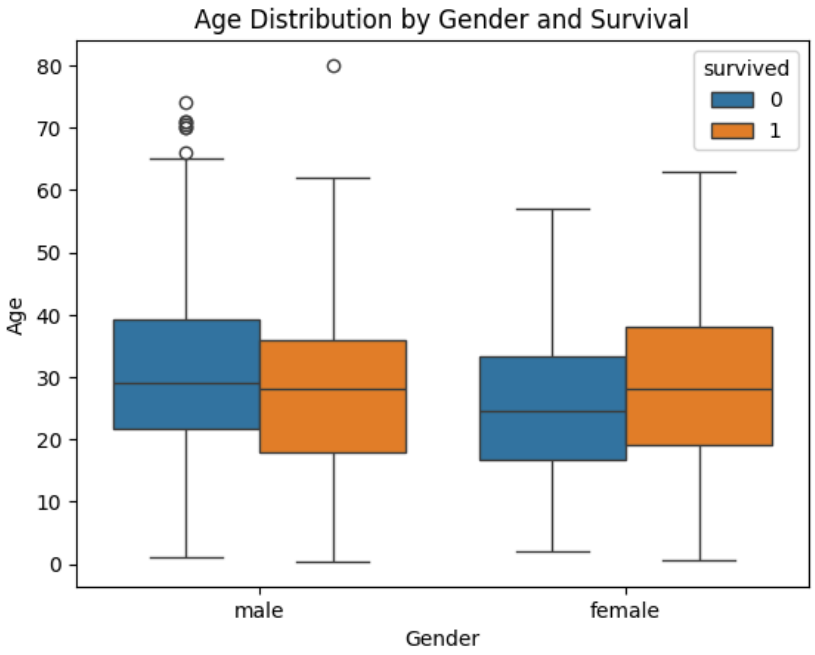
8. Difference between bar plot and histogram?

Bar Plot Histogram

Categorical Continuous

Gaps present No gaps

```
# Create box plot
sns.boxplot(x='sex', y='age', hue='survived', data=df)

# Add title and labels
plt.title("Age Distribution by Gender and Survival")
plt.xlabel("Gender")
plt.ylabel("Age")

# Show plot
plt.show()
```



What This Plot Shows

X-axis → Gender (sex)

Y-axis → Age (age)

Hue → Survival (0 = No, 1 = Yes)

Each box:

Middle line → median age

Box → 25% to 75% range

Whiskers → spread of data

Dots → outliers

1. What is a box plot?

A plot showing distribution using quartiles and median

2. What does the middle line represent?

Median (50th percentile)

3. What are whiskers in box plot?

Range of data excluding outliers

4. What are outliers?

Values far away from normal range

5. What did you infer from the plot?

Females had higher survival, age also influenced survival

6. Why use box plot instead of histogram?

To compare distributions across categories

7. What does hue represent here?

Survival category (0 or 1)

8. What does spread of box indicate?

Variability in data

9. Can box plot show skewness?

Yes, based on median position and whiskers

11. What happens if many outliers exist?